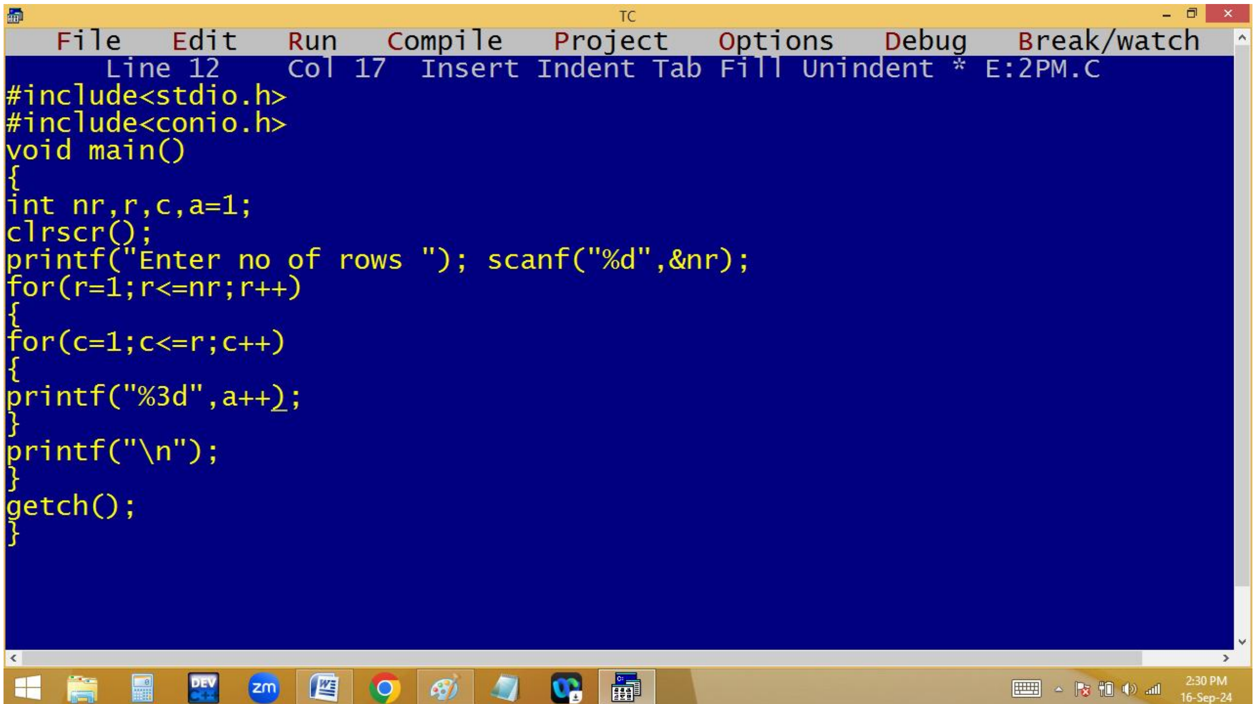
Floyd's triangle:

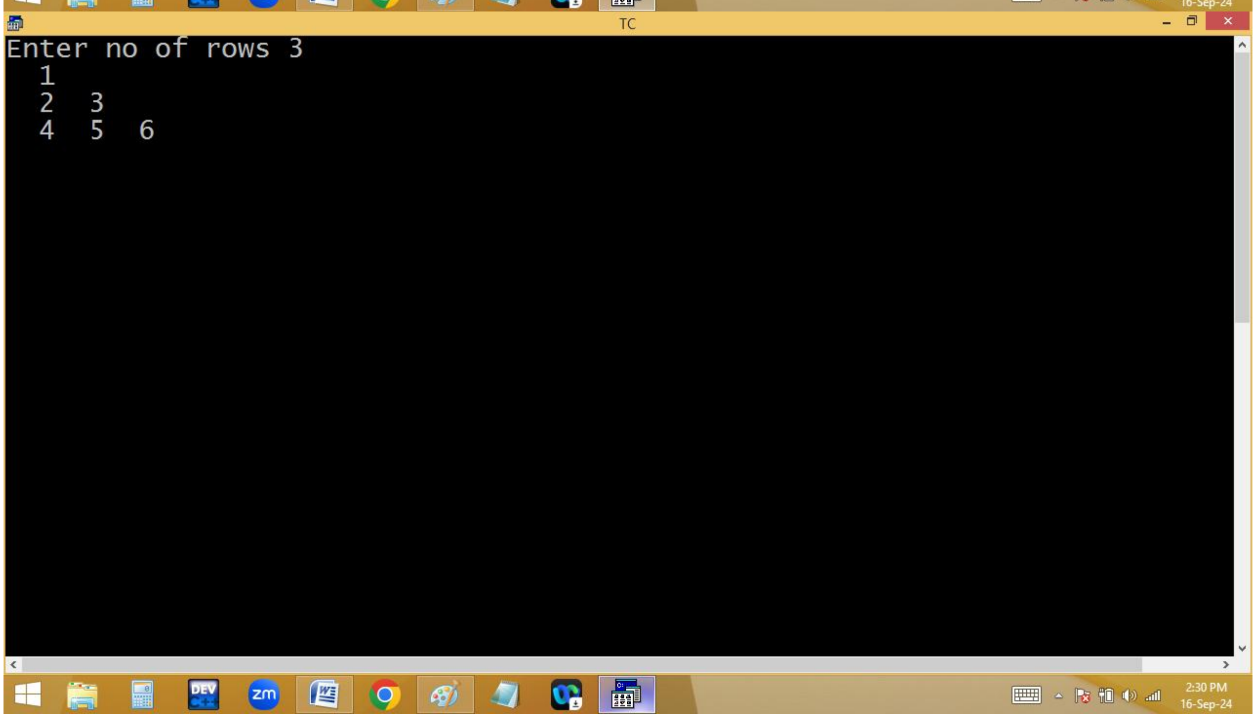
1

2 3

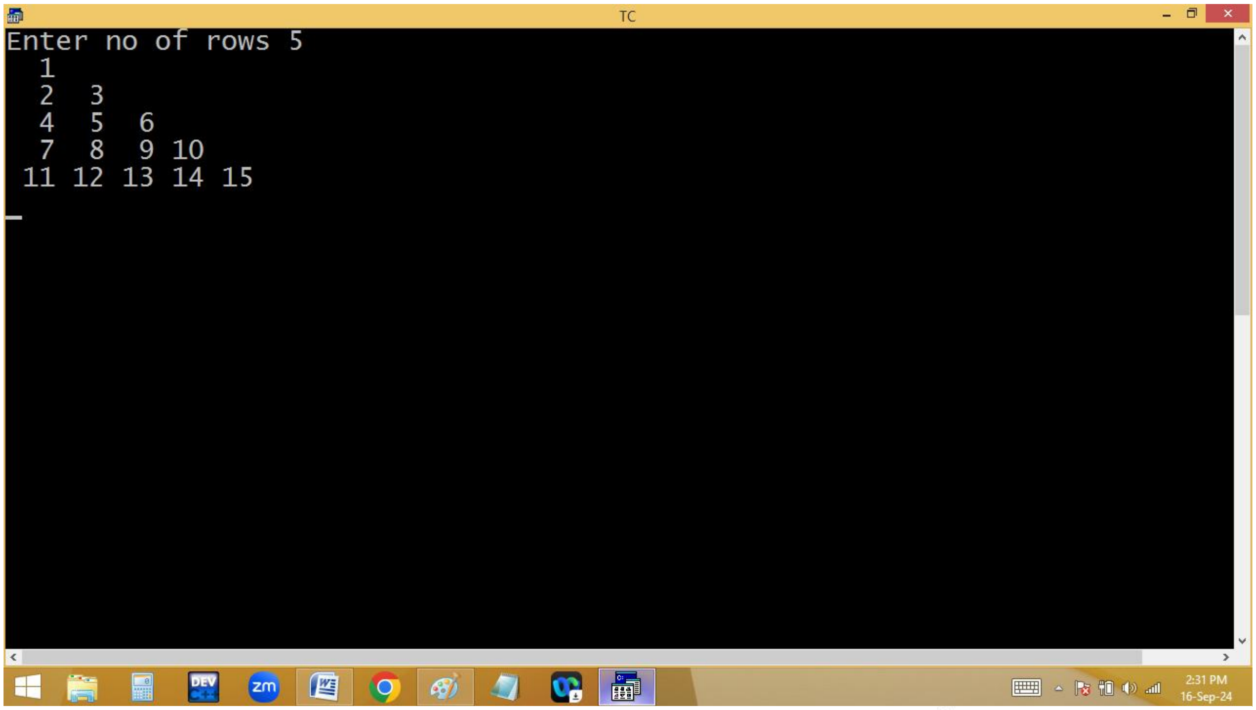
4 5 6



```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
    Line 12    Col 17  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a=1;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```



```
Enter no of rows 3
  1
  2  3
  4  5  6
```

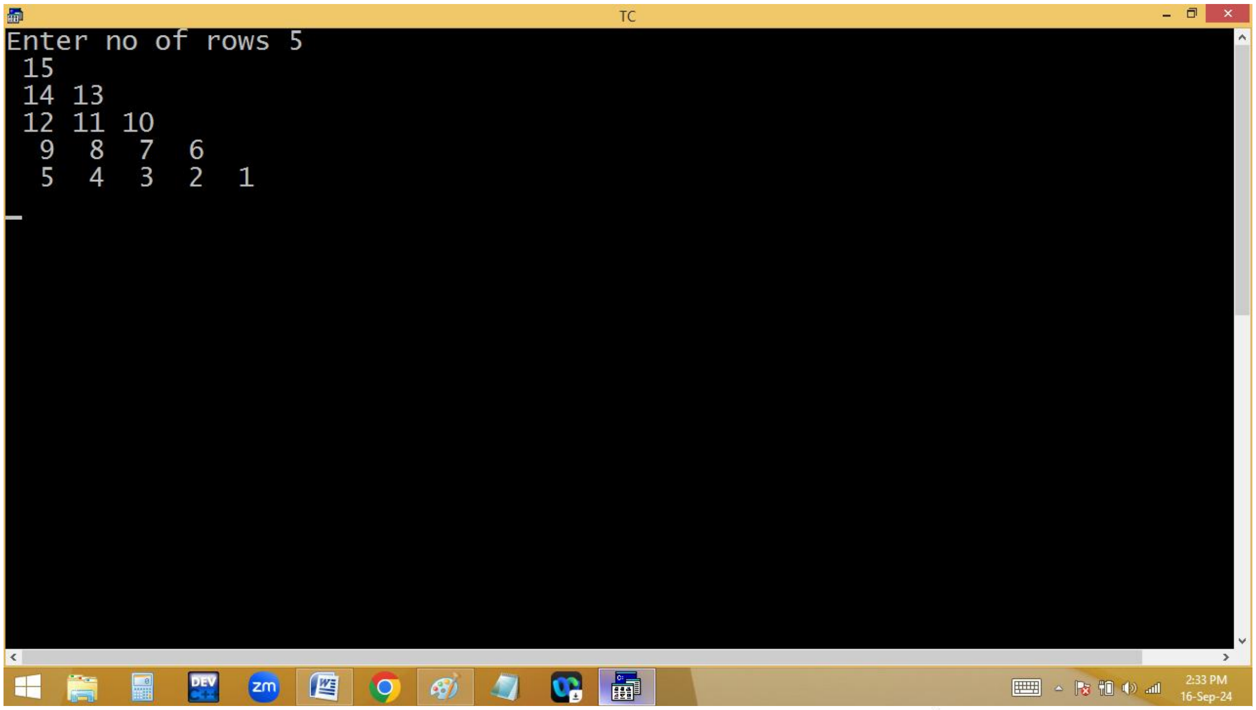


n	r	c=1 to r	
3	1	1 to 1	1
		1 to 2	2 3
		1 to 3	4 5 6

```
a=1;
for( r=1; r<=n; r++ )
{
for( c=1; c<=r; c++)
{
p(a++);
}
p("\n");
}
```

```
File  Edit  Run  Compile  Project  Options  Debug  Break/watch
     Line 7     Col 55  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr); a=nr*(nr+1)/2;
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("%3d",a--);
}
printf("\n");
}
getch();
}
```

```
Enter no of rows 3
  6
  5  4
  3  2  1
```



n	r	c=1 to r	
3	1	1 to 1	6
		1 to 2	5 4
		1 to 3	3 2 1

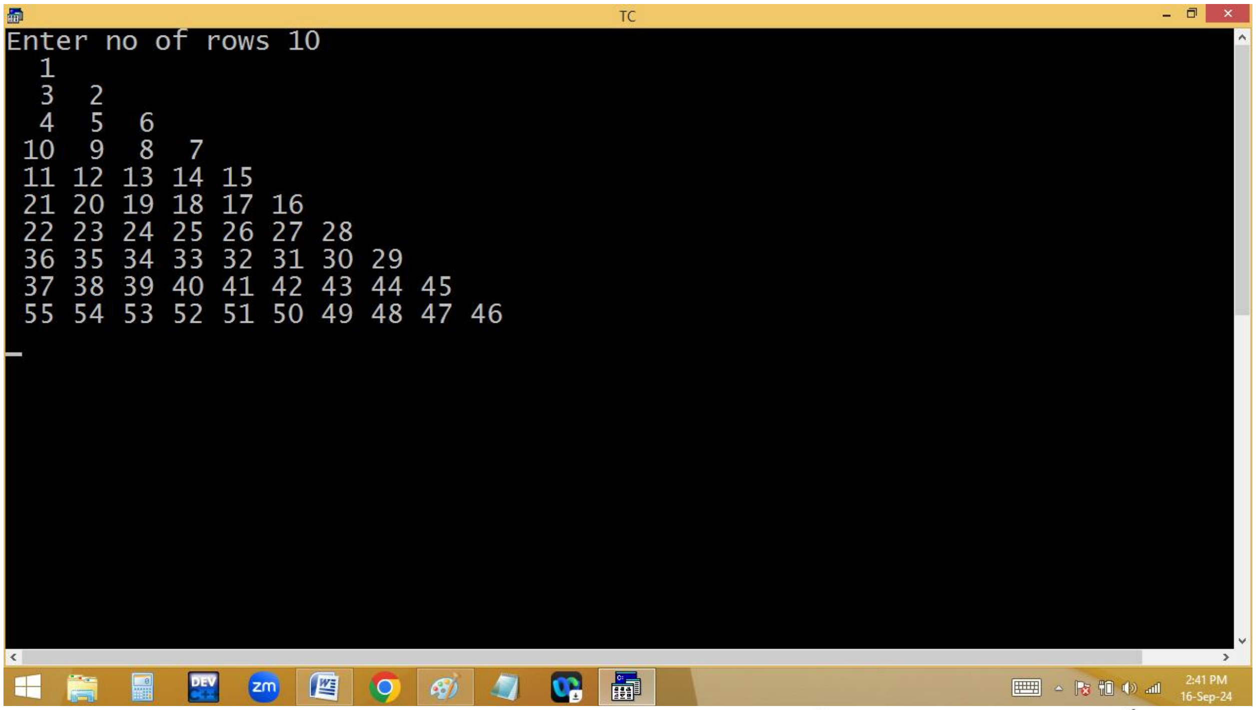
```
for( r=1; r<=n; r++ )
{
for( c=1; c<=r; c++)
{
p(a-- );
}
p("\n");
}
```

a = n*(n+1)/2; 3*4/2=6

a = 5*(6)/2=15

```
TC
  File    Edit    Run    Compile    Project    Options    Debug    Break/watch
       Line 9      Col 3    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a=1,b;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{ b=a+r-1;
for(c=1;c<=r;c++,a++)
{
if(r%2==0)printf("%3d",b--);else printf("%3d",a);
}
printf("\n");
}
getch();
}
```

```
TC
Enter no of rows 4
  1
  3  2
  4  5  6
 10  9  8  7
```

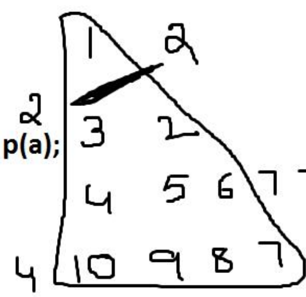


```
Enter no of rows 10
  1
  3  2
  4  5  6
 10  9  8  7
 11 12 13 14 15
 21 20 19 18 17 16
 22 23 24 25 26 27 28
 36 35 34 33 32 31 30 29
 37 38 39 40 41 42 43 44 45
 55 54 53 52 51 50 49 48 47 46
```

n	r	c=1 to r
3	1	1 to 1
		1 to 2
		1 to 3

1
2 - 3 2
4 5 6 7
4 - 10 9 8 7

```
for( r=1; r<=n; r++ )
{  b=a+r-1;       a++
for( c=1; c<=r; c++)
{
 if(r%2==0)p(b--);else p(a);
}
p("\n");
}
```



a++
1 2+2-1=3
3 4 5 6 7+4-1=10

TC

File Edit Run Compile Project Options Debug Break/watch

Line 10 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,a,b;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{ a=nr-1; b=a+r;
for(c=1;c<=r;c++)
{
if(c==1)printf("%3d",r);else {printf("%3d",b);a--;b=b+a;}
}
printf("\n");
}
getch();
}
```

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TC

```
Enter no of rows 10
  1
  2 11
  3 12 20
  4 13 21 28
  5 14 22 29 35
  6 15 23 30 36 41
  7 16 24 31 37 42 46
  8 17 25 32 38 43 47 50
  9 18 26 33 39 44 48 51 53
 10 19 27 34 40 45 49 52 54 55
```

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```
for( r=1; r<=n; r++ )
{   a=nr-1; b=a+r;
for( c=1; c<=r; c++)
{
if(c==1)p(r); else p(b);a--; b=b+a;
}
p("\n");
}
```

nr	r	c=1 to r
4	1	1 to 1
		1 to 2
		1 to 3

3

a = nr - 1 ==> 4-1=3 → 2 1

b = a+r ==> 3 + 2 = 5

3+3 = 6

4+3 = 7

1

2 5

3 6 8

4 7 9 10

2+3=5

3+3 = 6+2 = 8

4+3 = 7+2 = 9+1 = 10


```
for( r=1; r<=n; r++ )
{   a=nr-1; b=a+r;
for( c=1; c<=r; c++)
{
if(c==1)p(r); else p(b);a--; b=b+a;
}
p("\n");
}
```


nr	r	c	a=nr-1	b=a+r
4	1	1	4-1=3	3+2=5
	2	1	2	3+3=6+2=8
	3	1		4+3=7+1=9
	4	1		1

3
2
1

10

1

2 5

3 6 8

4 7 9 10

TC

File Edit Run Compile Project Options Debug Break/watch

Line 1 Col 33 Insert Indent Tab Fill Unindent * E:2PM.C

```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
if((r+c)%2==0)printf("$ ");else printf("* ");
}
printf("\n");
}
getch();
}
```

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TC

```
Enter no of rows 10
$
* $
$ * $
* $ * $
$ * $ * $
* $ * $ * $
$ * $ * $ * $
* $ * $ * $ * $
$ * $ * $ * $ * $
* $ * $ * $ * $ * $
```

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$			
1+1=2			
*	$		
2+1=3	2+2		
$	*	$	
3+1	3+2	3+3	
*	$	*	$

TC

File Edit Run Compile Project Options Debug Break/watch

Line 13 Col 19 Insert Indent Tab Fill Unindent * E:2PM.C

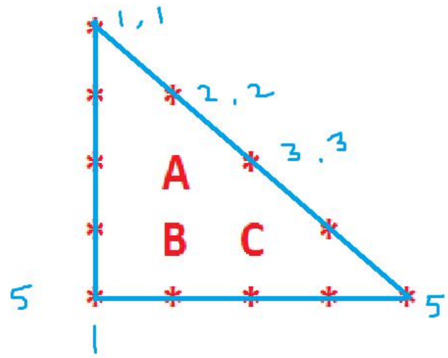
```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c; char ch='A';
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
if(c==1||r==c||nr==r)printf("* ");else printf("%c ",ch++);
if(ch>'Z') ch='A';_
}
printf("\n");
}
getch();
}
```

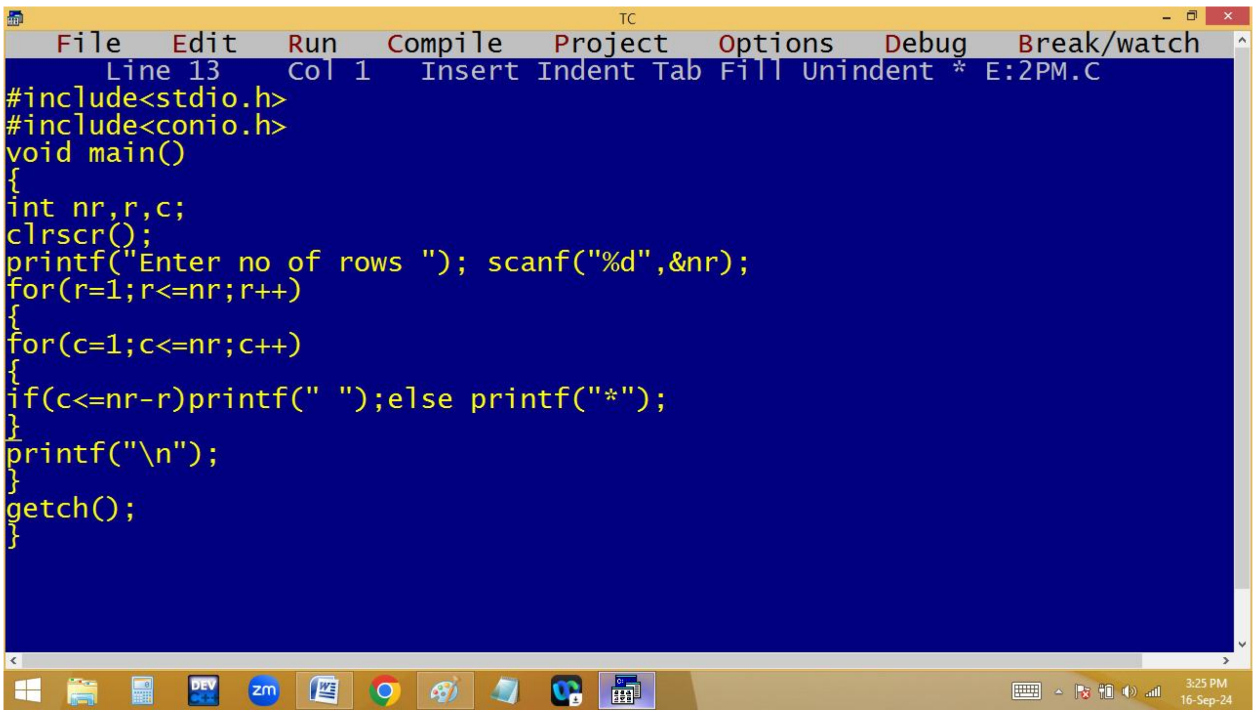
TC

```
Enter no of rows 15
*
* *
* A *
* B C *
* D E F *
* G H I J *
* K L M N O *
* P Q R S T U *
* V W X Y Z A B *
* C D E F G H I J *
* K L M N O P Q R S *
* T U V W X Y Z A B C *
* D E F G H I J K L M N *
* O P Q R S T U V W X Y Z *
* * * * * * * * * * * * * * *
```

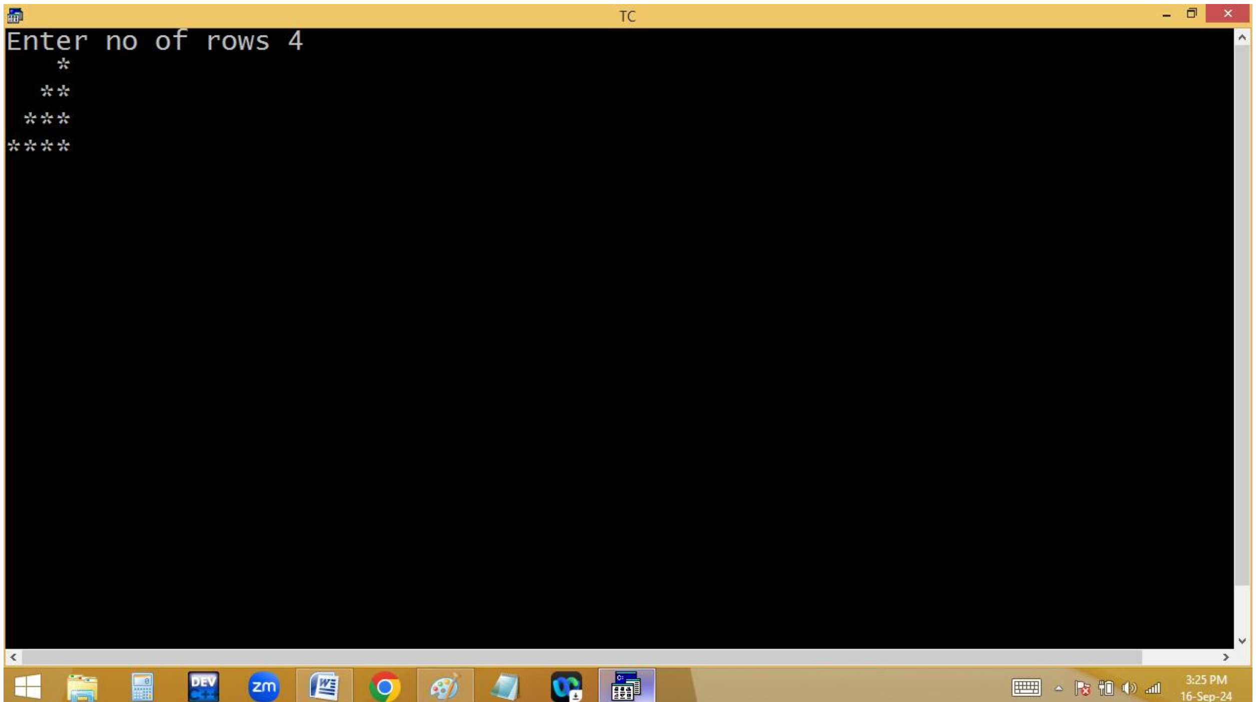
```
ch='A';

for( r=1; r<=n; r++ )
{
for( c=1; c<=r; c++)
{
 if(c==1 || c==r || n==r ) p(*); else p(ch++);
 }
p("\n");
}
```






```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nr;c++)
{
if(c<=nr-r)printf(" ");else printf("*");
}
printf("\n");
}
getch();
}
```


TC
Enter no of rows 4
   *
  **
 ***
****
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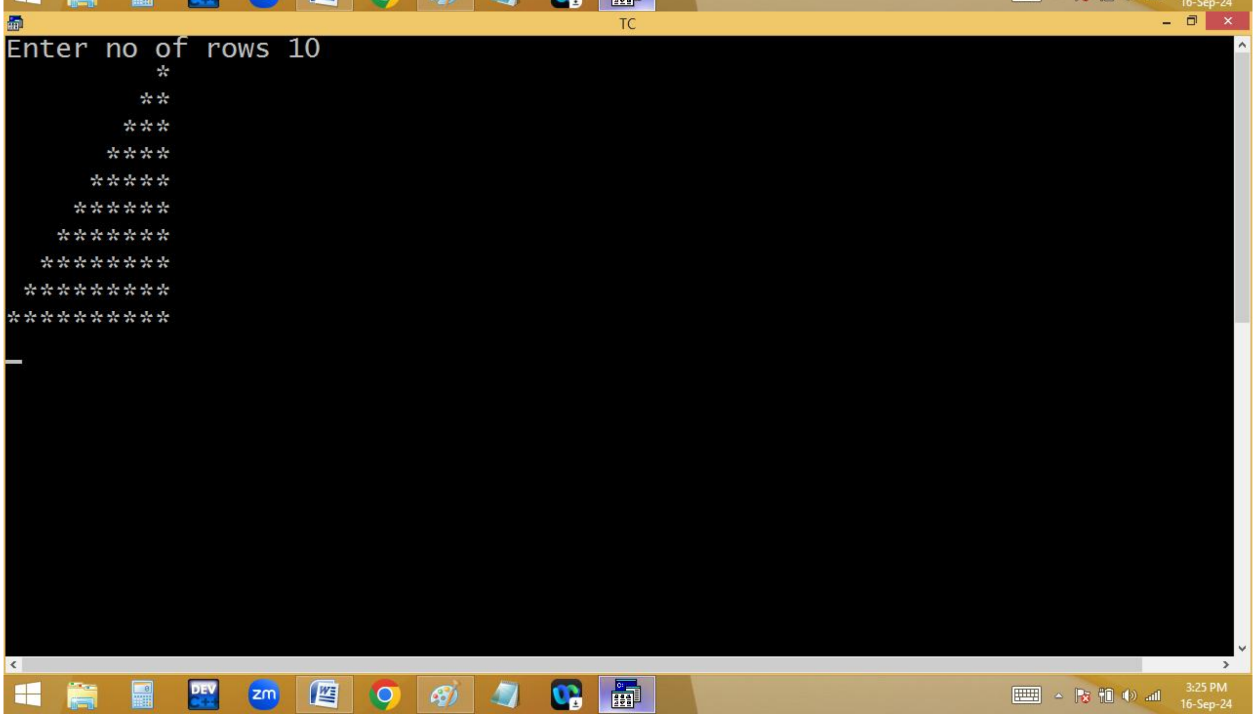
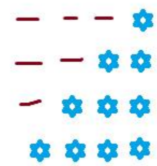
16-Sep-24

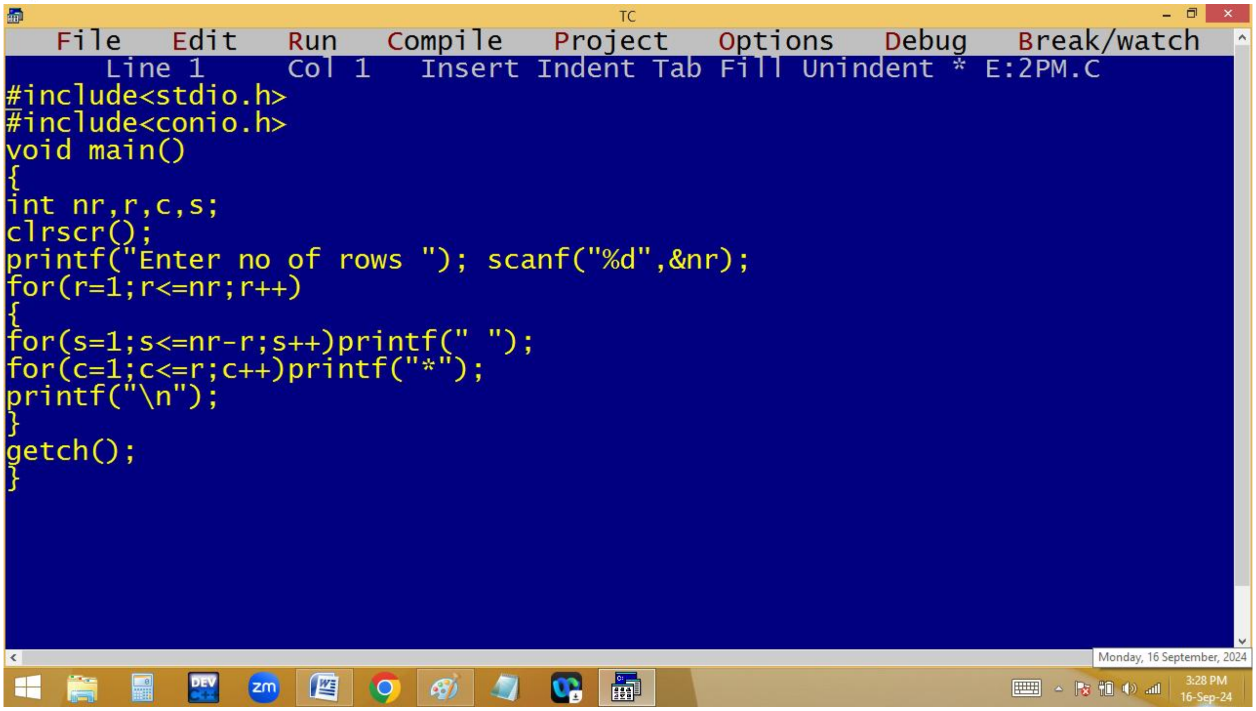

TC
Enter no of rows 10
         *
        **
       ***
      ****
     *****
    ******
   ****
  ****
 ****
****
3:25 PM
16-Sep-24

```
for( r=1; r<=nr; r++ )
{
for( c=1; c<=nr; c++)
{
if(c<=nr-1) p(" ");else p(*);
}
p("\n");
}
```

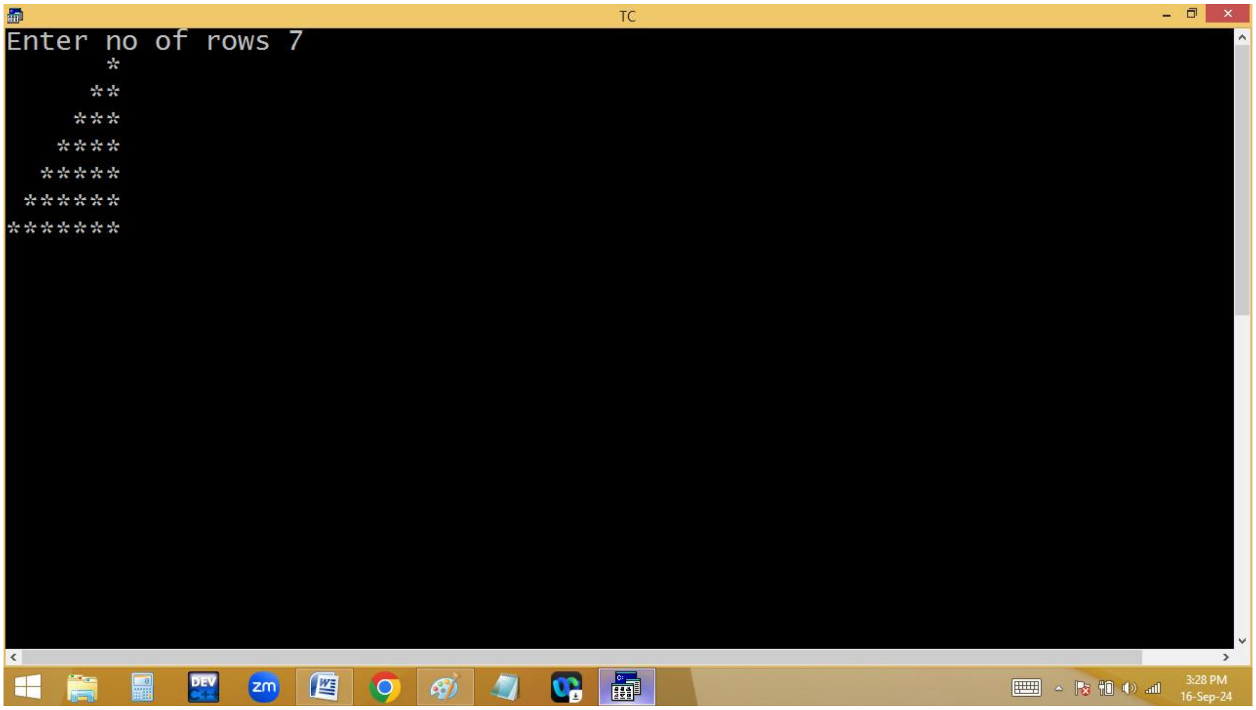
1<= 3 _ _ _ *
2 _ _ * *
1 _ * * *
0 * * * *

nr - r = s	c=1 to r	*
4 - 1 : 3	1 to 1 =	1
4 - 2 = 2	1 to 2 :	2
4 - 3 = 1	1 to 3 =	3
4 - 4 = 0	1 to 4 :	4





```
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)printf("*");
printf("\n");
}
getch();
}
```




```
for( r=1; r<=nr; r++ )
{           4-3
            4-2
            4-1
 for(s=1;s<=nr-r;s++)p(" ");
 for(c=1;c<=r;c++)p(*);
 p("\n");
 }
```

nr - r = s	c=1 to r	*
4 - 1 : 3	1 to 1 =	1
4 - 2 = 2	1 to 2 :	2
4 - 3 = 1	1 to 3 :	3
4 - 4 = 0	1 to 4 :	4

```
- - - *
- - * *
- * * *
* * * *
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 11 Col 28 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)printf("* ");
printf("\n");
}
getch();
}
```

TC

```
Enter no of rows 10
         *
        * *
       * * *
      * * * *
     * * * * *
    * * * * * *
   * * * * * * *
  * * * * * * * *
 * * * * * * * * *
* * * * * * * * * *
```

```
for( r=1; r<=nr; r++ )
{        4-3
         4-2
         4-1
for(s=1;s<=nr-r;s++)p(" ");
for(c=1;c<=r;c++)p(*   );
p("\n");
}
```

nr - r = s	c=1 to r	*
4 - 1 : 3	1 to 1 =	1
4 - 2 = 2	1 to 2 :	2
4 - 3 = 1	1 to 3 =	3
4 - 4 = 0	1 to 4 :	4

```
      *
    *   *
  *   *   *
*   *   *   *
```

```
   File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 12     Col 12  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)printf("**");
printf("\b \n");
}
getch();
}
```

```
Enter no of rows 10
         *
        ***
       *****
      *******
     *********
    ***********
   *************
  ***************
 *****************
*******************
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 11 Col 17 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)printf("*");
printf("\n");
}
getch();
}
```

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TC

```
Enter no of rows 10
         *
        ***
       *****
      *******
     *********
    ***********
   *************
  ***************
 *****************
*******************
```

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```
File  Edit  Run  Compile  Project  Options  Debug  Break/watch
      Line 11    Col 55  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=r;c++)if(c<r)printf("**");else printf("*");_
printf("\n");
}
getch();
}
```

```
Enter no of rows 10
         *
        ***
       *****
      *******
     *********
    ***********
   *************
  ***************
 *****************
*******************
```

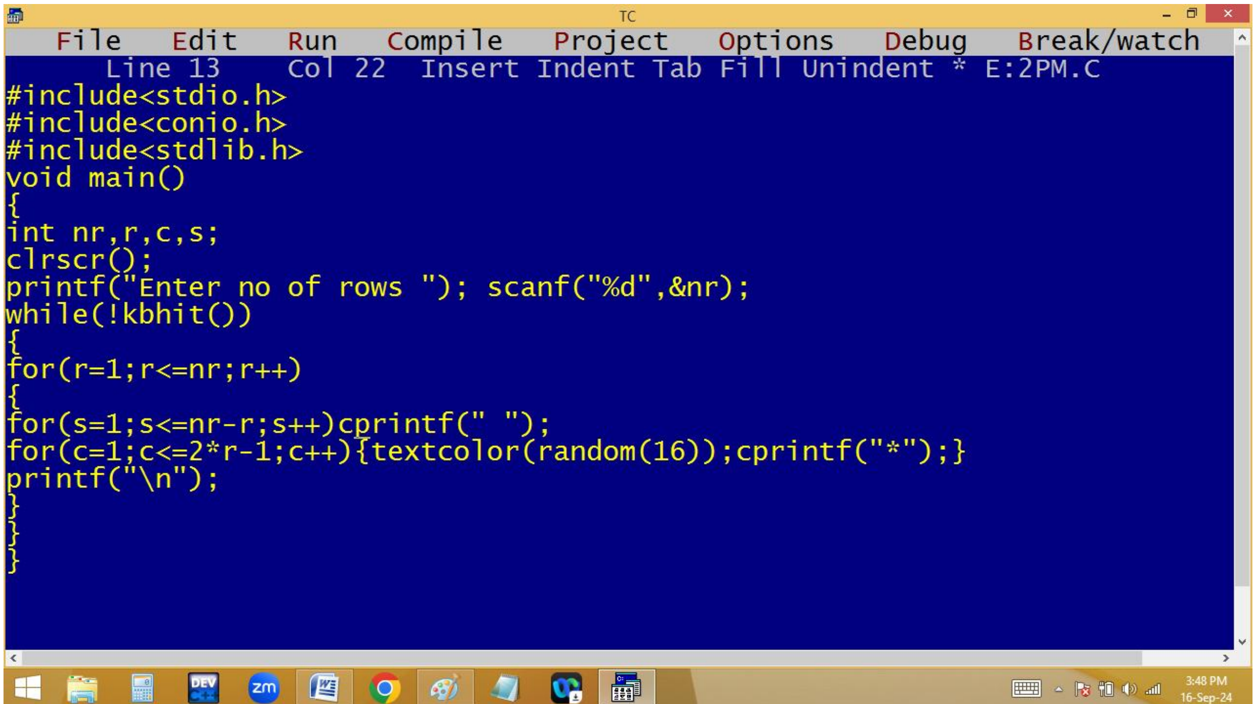
```
for( r=1; r<=nr; r++ )
{            4-3
             4-2
             4-1
for(s=1;s<=nr-r;s++)p(" ");
for(c=1;c<=r;c++)p(*  );
p("\n");
}
```

nr - r = s	c=1 to r	*
4 - 1 = 3	1 to 1 =	1
4 - 2 = 2	1 to 2 =	2
4 - 3 = 1	1 to 3 =	3
4 - 4 = 0	1 to 4 =	4

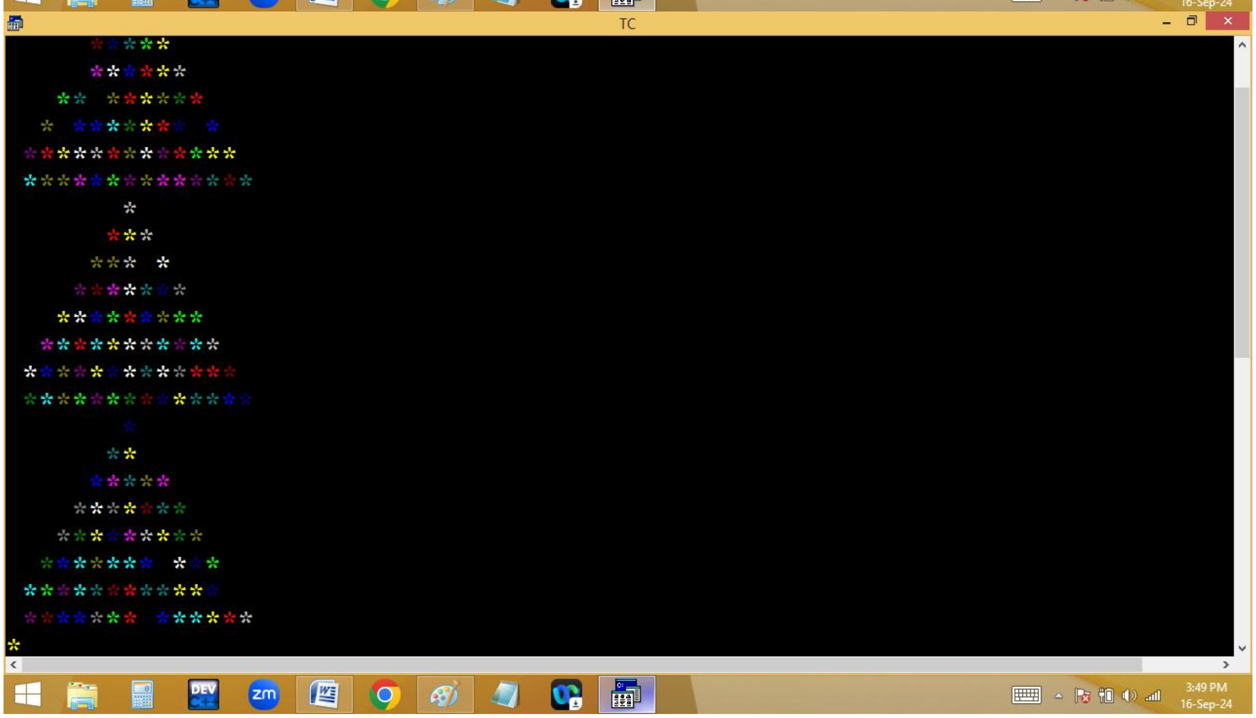
```
      *
    * * *
  * * * * *
* * * * * * *
```

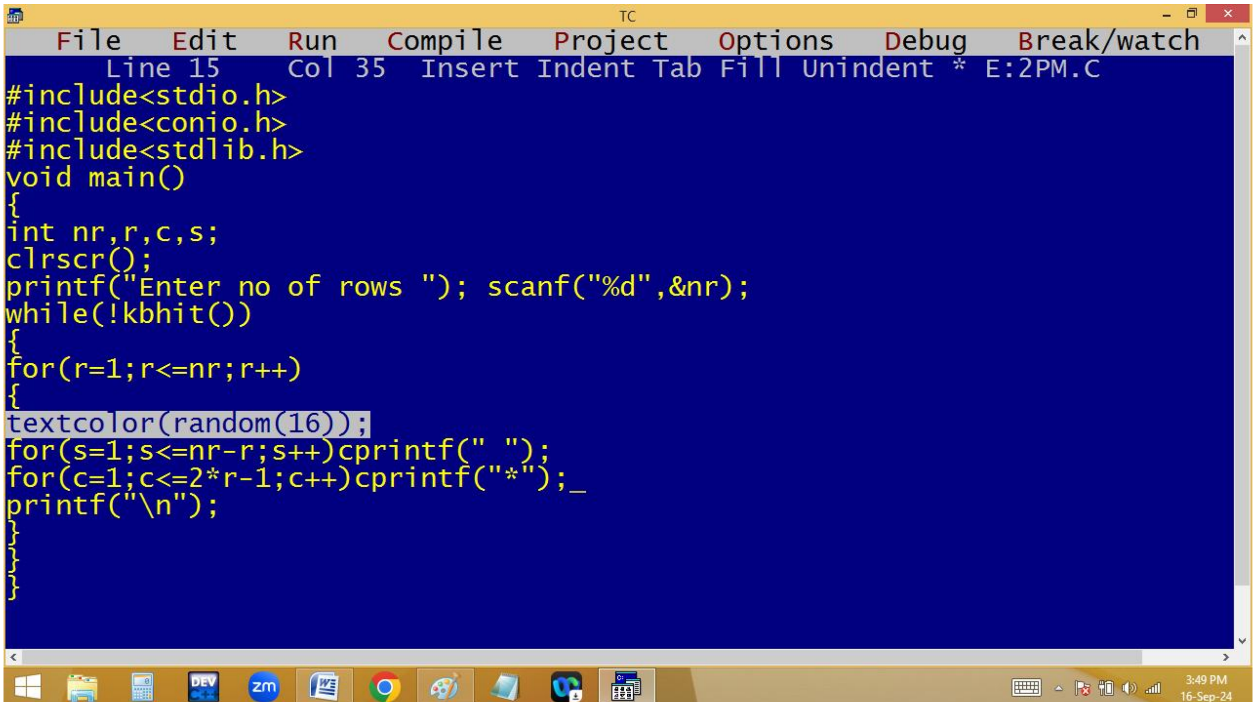
* *   * *

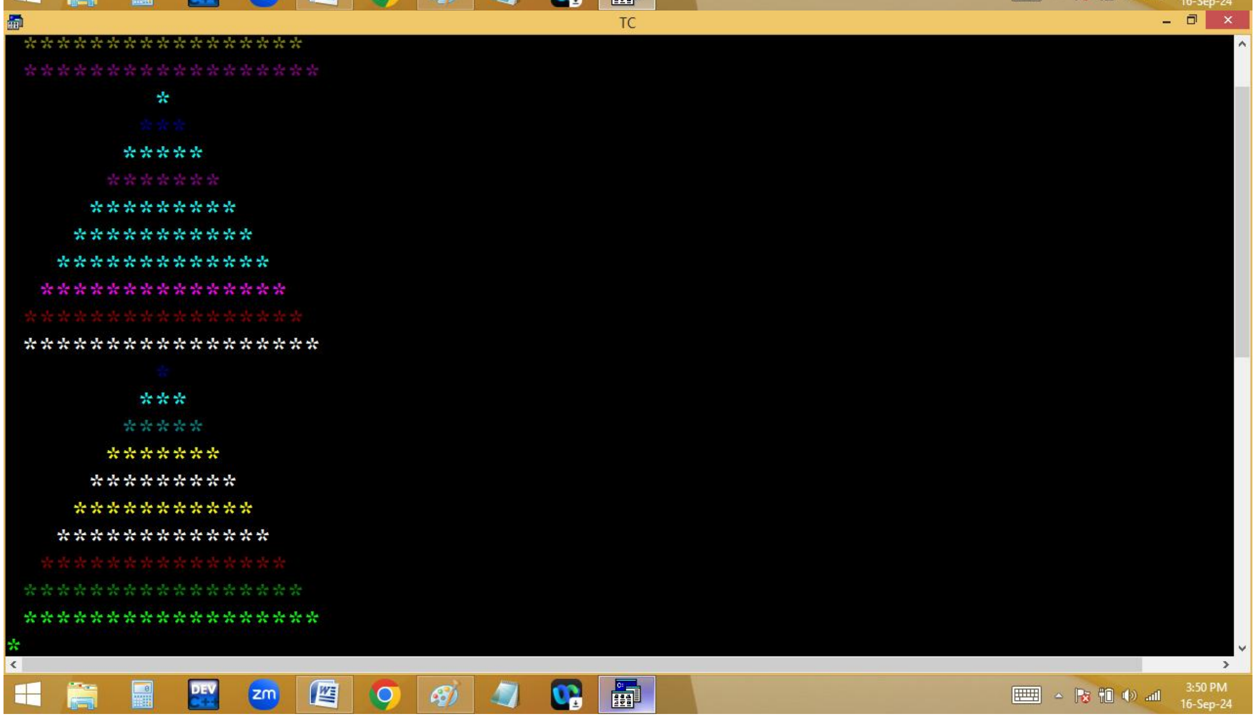
r	c
2	1< 2 * * *
	2==2
3	1<3 * * * * *
	2<3
	3==3




```c
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++){textcolor(random(16));cprintf("*");}
printf("\n");
}
}
}
```




TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 35 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
for(r=1;r<=nr;r++)
{
textcolor(random(16));
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");_
printf("\n");
}
}
}
3:49 PM
16-Sep-24


TC
****
****
*
***
*****
****
****
****
****
****
****
****
*
***
*****
****
****
****
****
****
****
****
*
3:50 PM
16-Sep-24

```
  File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 14     Col 1    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
textcolor(random(16));
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");
printf("\n");
}
}
}
```

```
 *****************
 *******************
          *
         ***
        *****
       *******
      *********
     ***********
    *************
   ***************
  *****************
 *******************
          *
         ***
        *****
       *******
      *********
     ***********
    *************
   ***************
  *****************
 *******************
*
```

```
TC
  File   Edit   Run   Compile   Project   Options   Debug   Break/watch
      Line 11     Col 20   Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
textcolor(LIGHTRED);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");
printf("\n");
}
}
}
```

```
TC
 *******
 ********
    *
   ***
  *****
 *******
 ********
    *
   ***
  *****
 *******
 ********
    *
   ***
  *****
 *******
 ********
    *
   ***
  *****
 *******
 ********
*
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
while(!kbhit())
{
textcolor(LIGHTRED);
for(r=nr;r>=1;r--)
{
for(s=1;s<=nr-r;s++)cprintf(" ");
for(c=1;c<=2*r-1;c++)cprintf("*");
printf("\n");
}
}
}
```

TC

```
  ***
   *
********
*******
 *****
  ***
   *
********
*******
 *****
  ***
   *
********
*******
 *****
  ***
   *
********
*******
 *****
  ***
   *
```

TC

File Edit Run Compile Project Options Debug Break/watch

Line 1 Col 37 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h> #include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)printf("*");
printf("\n");
}
for(r=nr-1;r>=1;r--)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)printf("*");
printf("\n");
}
getch();
}
```

3:55 PM 16-Sep-24

TC

```
Enter no of rows 5
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

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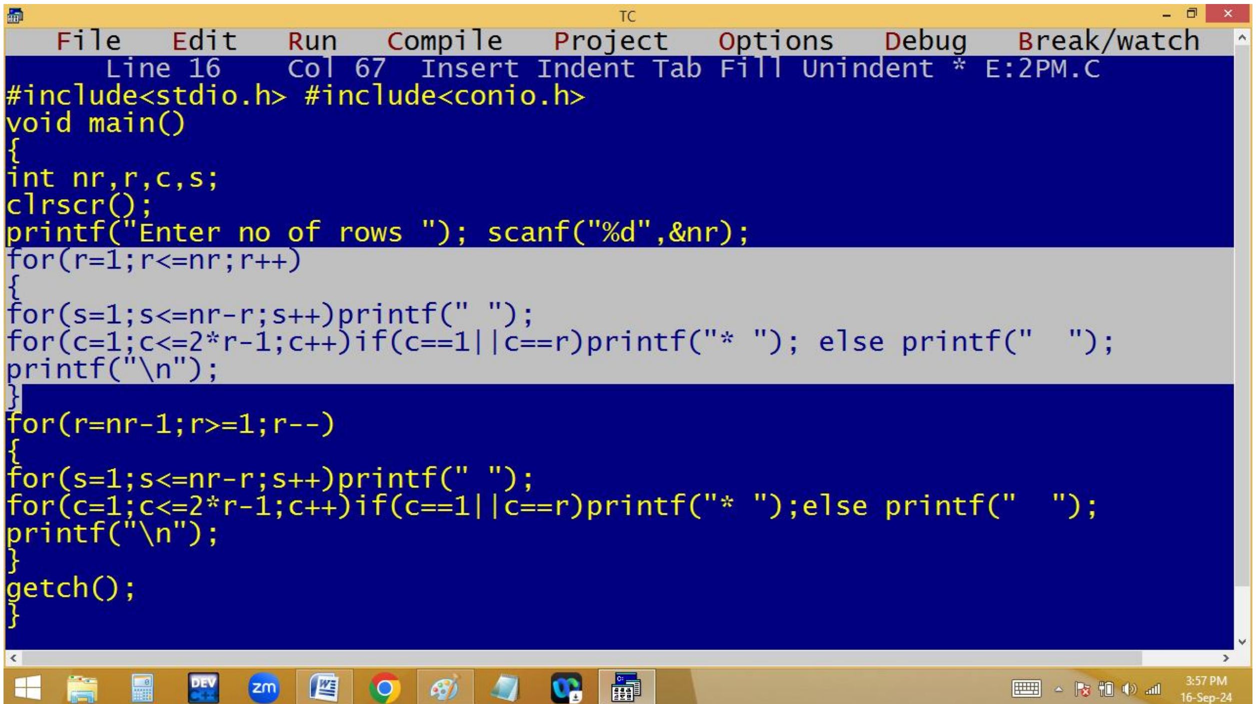
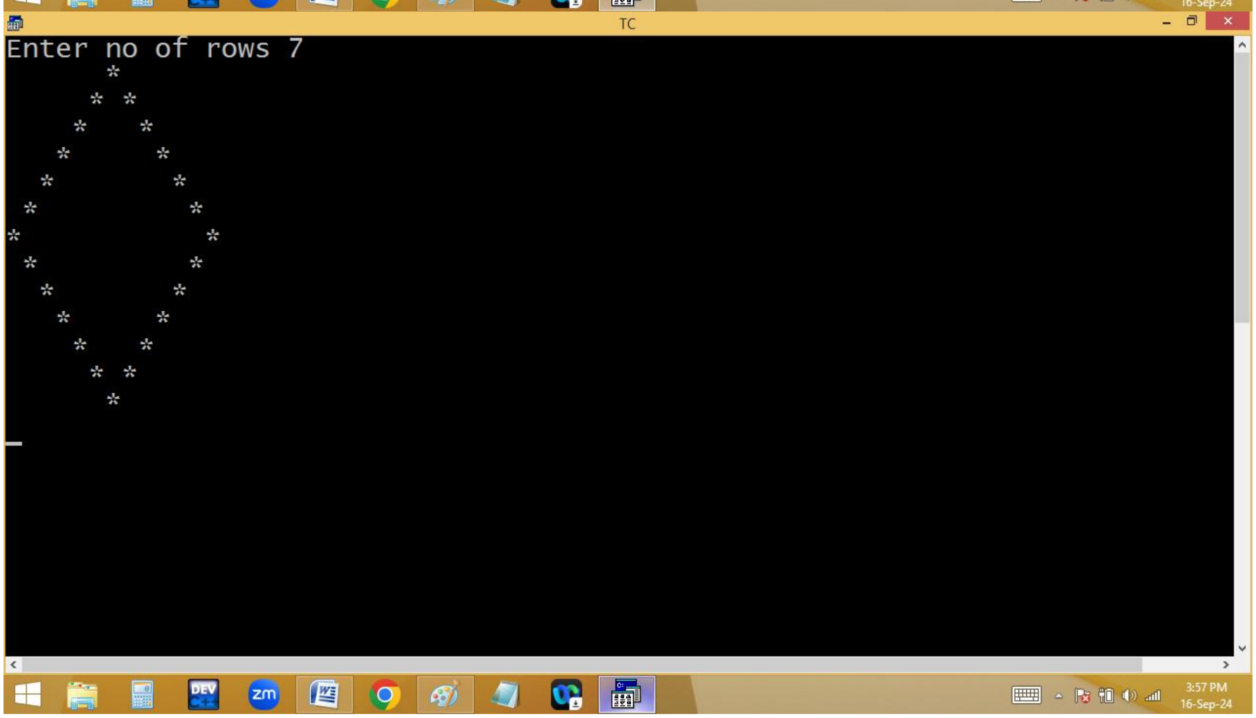



```
#include<stdio.h> #include<conio.h>
void main()
{
int nr,r,c,s;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)if(c==1||c==r)printf("* "); else printf("  ");
printf("\n");
}
for(r=nr-1;r>=1;r--)
{
for(s=1;s<=nr-r;s++)printf(" ");
for(c=1;c<=2*r-1;c++)if(c==1||c==r)printf("* ");else printf("  ");
printf("\n");
}
getch();
}
```




```
Enter no of rows 7
      *
     * *
    *   *
   *     *
  *       *
 *         *
*           *
 *         *
  *       *
   *     *
    *   *
     * *
      *
```